

Computer Algebra (SCI19 3112) — Final Cheat Sheet

Groups/Euler/crypto · Complex/polar/roots · Iso · FTA/fields · Coeff/value · FFT/IFFT · GCD/systems

1 Groups Recap + Euler φ

Group: closure, identity e (unique, $aoe=a$), associativity, inverses (unique). $(\mathbb{Z}, +)$ infinite group; $(\mathbb{Z}_n, +)$ finite, always cyclic (gen $[1]_n$); $\mathbb{Z}_n^*$ group $\iff n$ prime. $|G|=\#\text{elem}$; $\text{ord}(g)$ = smallest $n>0$ with $g^n=e$.

Lagrange: $\text{ord}(g) \mid |G|$, $g^{|G|}=e$

Cyclic: $G=\langle g \rangle \iff \text{ord}(g)=|G|$

Worked: $|\mathbb{Z}_8^+|=8$; $\text{ord}[1]_8=8$; $\text{ord}[2]_8=4$ (2, 4, 6, 0); $\text{ord}[7]_8=8$ (7, 6, 5, 4, 3, 2, 1, 0); gens $[1]_8, [3]_8, [5]_8, [7]_8$.

$\varphi(n)=\#\{1 \leq k \leq n : \gcd(k,n)=1\}$. **Formulas:** p prime

$\varphi(p)=p-1$; $\varphi(p^k)=p^k-p^{k-1}$; coprime

$\varphi(mn)=\varphi(m)\varphi(n)$.

Worked: $\varphi(16)=8$ (1, 3, 5, 7, 9, 11, 13, 15).

$\varphi(17)=16$

$\varphi(14)=\varphi(2)\varphi(7)=(2-1)(7-1)=1 \cdot 6=6$

$\varphi(9)=9-3=6$ (prime power)

Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a,n)=1$ (= Lagrange on units mod n , $|G|=\varphi(n)$). FLT = prime case ($\varphi(p)=p-1$) = Lagrange on $\mathbb{Z}_p^*$. **Use:** prime $\rightarrow$ FLT (smaller exponent); composite $\rightarrow$ Euler. **Idea:** reduce exponent mod $\varphi(n)$.

Worked $3^{347} \bmod 14$: Given $\gcd(3,14)=1$, $\varphi(14)=6$.

Step 1 (reduce exponent): $347=6 \cdot 57+5$ ($6 \cdot 57=342$).

Step 2 (evaluate): $3^5=243$;

$243-17 \cdot 14=243-238=5 \Rightarrow [5]_{14}$. **Check:** $\text{pow}=5$.

2 Crypto

Caesar: $c \equiv m+s \pmod{26}$, $m \equiv c-s$. **Worked**

BY (B=1, Y=24), $s=20$: $1+20=21=\text{V}$,

$24+20=44-26=18=\text{S} \Rightarrow \text{VS}$.

Power: $c \equiv m^b \pmod{p}$; $m \equiv c^d$, $bd \equiv 1 \pmod{p-1}$. **Worked** $p=29, b=5$: B: $1^5=1=\text{B}$.

Y Step 1 ($24 \equiv -5$): $(-5)^2=25$; **Step 2:**

$25^2=625$; $625-21 \cdot 29=625-609=16$.

Y Step 3: $16 \cdot (-5)=-80$;

$-80+3 \cdot 29=-80+87=7 \Rightarrow \text{H}$. d (test d or Euclid):

$5d \equiv 1 \pmod{28} \Rightarrow d=17$ ($5 \cdot 17=85=3 \cdot 28+1$).

Correctness = bijective: $D(E(x))=x$,

$E(D(y))=y$. Caesar always bijective (shift). Power

bijective $\iff \gcd(b, p-1)=1$ (else many-to-one).

Many-to-one: ≥ 2 symbols map together;

surjective: covers whole target.

Check decrypt: VS-20: $21-20=1=\text{B}$,

$18-20=-2+26=24=\text{Y}$. Power: $7^{17} \bmod 29=24=\text{Y}$

Check OK.

3 Complex Numbers

Rect $z=a+bi$ ($\text{Re}=a$, $\text{Im}=b$), plane vector (a, b) .

Polar $r=\cos \theta+i \sin \theta$, $r=|z|=\sqrt{a^2+b^2}$. $|z|^2=z\bar{z}$,

$\bar{z}=a-bi$.

$+$, $-$ = vector + (parallelogram). Mult:

$(a+bi)(c+di)=(ac-bd)+(ad+bc)i$.

Worked mult: $(1+i)(2-3i)=(2+3)+(2-3)i=5-i$.

Conj rule: $\bar{z_1 z_2}=\bar{z_1} \bar{z_2}$: $(1-i)(2+3i)=5+i=5-i$.

Modulus + div (Quiz6): $|z_1||z_2|=\sqrt{2} \cdot \sqrt{13}=\sqrt{26}$;

$z_1/z_2=(1+i)(2+3i)/(4+9)=(-1+5i)/13$.

Worked 3+4i: $|z|=5$,

$z^{-1}=(3-4i)/25=0.12-0.16i$.

Worked $(1+6i)/(1-6i)$:

$(1+6i)^2/37=(-35+12i)/37$.

$\approx -0.9459+0.3243i$ **Check** ($37 \cdot (-0.9459)=-35$,

$37 \cdot 0.3243=12$).

Euler: $e^{i\theta}=\cos \theta+i \sin \theta$; **DeMoivre:**

$(re^{i\theta})^n=r^n(\cos n\theta+i \sin n\theta)$.

n -th roots: $\sqrt[n]{r} \left(\cos \frac{\theta+2k\pi}{n} + i \sin \frac{\theta+2k\pi}{n} \right)$,

$k=0..n-1$.

Div: $z_1/z_2=z_1\bar{z_2}/|z_2|^2$; **Inverse:** $z^{-1}=\bar{z}/|z|^2$.

Polar mult: $r_1 r_2 e^{i(\theta_1+\theta_2)}$;

$e^{i\pi/3} \cdot e^{i\pi/6}=e^{i\pi/2}=i$.

$\sin(-\theta)=-\sin \theta$, $\cos(-\theta)=\cos \theta$. **Use (Quiz6**

roots): $\sqrt{2} e^{i\pi/6}=\sqrt{2} \left(\frac{\sqrt{3}}{2} + \frac{i}{2} \right)=\frac{\sqrt{3}}{\sqrt{2}} + \frac{i}{\sqrt{2}}$.

Worked $8e^{i\pi/3}$: $8(\frac{1}{2}+i\frac{\sqrt{3}}{2})=4+4\sqrt{3}i \approx 4+6.9282i$.

Worked $11e^{3\pi i/2}$: $11(0-i)=-11i$.

Cube roots of $e^{i\pi/2}$: angles

$\pi/6, 5\pi/6, 3\pi/2 \Rightarrow \sqrt{3}/2+i/2, -\sqrt{3}/2+i/2, -i$.

Square roots of $e^{i\pi/4}$: angles $\pi/8, 9\pi/8$:

$\cos \frac{\pi}{8}=\sqrt{2+\sqrt{2}}/2 \approx 0.9239$,

$\sin \frac{\pi}{8}=\sqrt{2-\sqrt{2}}/2 \approx 0.3827$.

4th roots of i : angles $\pi/8, 5\pi/8, 9\pi/8, 13\pi/8$:

$0.9239+0.3827i, -0.3827+0.9239i$,

$-0.9239-0.3827i, 0.3827-0.9239i$ (**Check**

$\text{first}^2=e^{i\pi/4}$). **Rect $\rightarrow$ polar:** $2+2i$: $r=\sqrt{8}=2\sqrt{2}$,

$\theta=\pi/4 \Rightarrow 2\sqrt{2} e^{i\pi/4}$ (**Check**

$2\sqrt{2}(\sqrt{2}/2+i\sqrt{2}/2)=2+2i$).

Geometry: $\times e^{i\theta}$ rotates by θ (lengths mult); $\bar{z}$

mirrors across real axis; unit circle $\{|z|=1\}$

subgroup (closed $|zw|=1$, inv $\bar{z}$). **Plot recipe:**

mark Re on x , Im on $y \rightarrow$ arrow; add tip-to-tail;

mult adds angles, mults lengths.

Groups: $(\mathbb{C}, +)$ $e=0$ inv $-z$; $(\mathbb{C}^*, \cdot)$ $e=1$ inv $\bar{z}/|z|^2$;

$C_4^*=\{\pm 1, \pm i\}$ gen i ; C_8^* gen $(1+i)/\sqrt{2}$; $\{1, -1\}$

group (gen -1); $\{i, -i\}$ **not** ($i \cdot i=-1$ missing).

Mult prop $|z_1 z_2|=|z_1||z_2|$: $z_1=(1+i)/\sqrt{2}$,

$z_2=(1-i)/\sqrt{2}$: $1 \cdot 1=1=1$ **Check OK**.

4 Roots of Unity

Powers $\omega_n^0 \dots \omega_n^{n-1}$ = all n roots cyclically (regular n -gon on unit circle, angle $2\pi/n$ apart); primitive = generator; n distinct roots of $z^n-1=0$.

Primitive: ω_n^k generates all $\iff \gcd(k, n)=1$;

e.g. $n=8$: $k=1, 3, 5, 7$. **Why:**

$\text{ord}(\omega_n^k)=n/\gcd(k, n)$ — shared factors shortcut the cycle (e.g. $\omega_8^2=\omega_4$, order 4).

Gen test: $g^d \neq 1$ for all proper divisors $d|n$ ($\mathbb{Z}_{13}^*$: $2^1, 2^2, 2^3, 2^4, 2^6=2$, 4, 8, 3, $12 \neq 1$).

Cyclic: $C_n^*=\langle \omega_n \rangle$; reduce first: $\omega_8^2=\omega_4=i$.

Root: $\omega_n=e^{2\pi i/n}$.

Relations (n even; last two $\forall n$): $\omega_n^n=1$, $\omega_n^{n/2}=-1$, $\omega_n^2=\omega_{n/2}$, $\omega_n^{-1}=\bar{\omega}_n$, $\omega_n^{k+n/2}=-\omega_n^k$, $\omega_n^{2k}=\omega_{n/2}^k$, $\omega_n^{n-k}=\bar{\omega}_n^k$.

Worked $\omega_8=(1+i)/\sqrt{2}$: $\omega_8^4=-1$, $\omega_8^8=1$,

$\omega_8^{-5}=\omega_8^{-8+3}=\omega_8^3=(-1+i)/\sqrt{2}$;

$\omega_{16}^8=e^{8\pi i/16}=i=\omega_4$. $\sqrt{i}=(1+i)/\sqrt{2}$ (**Check**

$((1+i)/\sqrt{2})^2=i$); $\sqrt{-i}=(-1+i)/\sqrt{2}$.

Eval trick: reduce exponents mod n , use $\omega_2=-1$.

Worked $p=x^4-3x^3+x^2-2x-1$ at $\omega_4=i$: powers $\omega_4^4=-1$, $\omega_4^3=-i$, $\omega_4^4=1$.

Then

$1-3(-i)+(-1)-2i-1=1+3i-1-2i-1=-1+i$.

Lecture: $3\omega_4^3-2\omega_4^2+2\omega_4^8=3(-i)+2+2+1=5-3i$.

Power drill: $[(\omega_8^3)^4-1]$: $\omega_8^{12}=\omega_4^3=-1$, so

$-1-1=-2$.

C3 (Quiz6): $\omega_3=\frac{-1+\sqrt{3}i}{2}=e^{2\pi i/3}$, $\omega_3^2=\frac{-1-\sqrt{3}i}{2}$, $\omega_3^3=1$; table via powers: $\omega_3^2 \cdot \omega_3^2=\omega_3^4=\omega_3$; gens $\omega_3, \omega_3^2 \leftrightarrow [1]_3, [2]_3$ ($\gcd=1$).

5 Iso $C_n^* \cong \mathbb{Z}_n^+$

Map $\omega_n^k \leftrightarrow [k]_n$; $\times \leftrightarrow +$:

$\omega_n^k \omega_n^l=\omega_n^{k+l} \leftrightarrow [k]_n+[l]_n=[k+l]_n$. Gen $\leftrightarrow$ gen.

Iso: bijection preserving op (C_4^* table $\equiv \mathbb{Z}_4^+$ table).

Subgroup: subset + closed under op.

Worked $C_4 \leftrightarrow \mathbb{Z}_4$:

$[0]_4 \leftrightarrow 1, [1]_4 \leftrightarrow i, [2]_4 \leftrightarrow -1, [3]_4 \leftrightarrow -i$;

$[1]_4+[2]_4=[3]_4 \leftrightarrow i+(-1)=-i$;

$[3]_4+[2]_4=[5]_4=[1]_4 \leftrightarrow (-i)+(-1)=i$. Tables same shape $\Rightarrow$ iso.

No-iso test: $|G|$ must match + generator orders;

e.g. $|C_2^*|=2$ vs $|\mathbb{Z}_4^+|=4 \Rightarrow$ no iso. **Fill C_5^* via iso:**

$\omega_5^3 \cdot \omega_5^4=\omega_5^7=\omega_5^2 \leftrightarrow [3]_5+[4]_5=[7]_5=[2]_5$. Subgroups: $C_{2m} \subset C_{2m+1}$ ($z^{2m}=1 \Rightarrow (z^2)^m=1$).

6 Poly: FTA, Fields, Coeff/Value

Field: $+$, $-$, $\times$, $/$ (no $/0$); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ yes; $\mathbb{Z}$ no (a/b

can leave $\mathbb{Z}$); $\mathbb{N}$ no ($a-b$ can leave $\mathbb{N}$); $\mathbb{Z}_p$ (p prime)

yes (every $\neq 0$ invertible). Polys over field K

(usually $\mathbb{R}/\mathbb{C}$); some fields give no roots.

Alg closed: every $\deg \geq 1$ has a root; $\mathbb{C}$ is (FTA); $\mathbb{R}$ is **not** ($x^2+1=0$ no real root).

FTA: $\deg n \Rightarrow$ **exactly n roots in $\mathbb{C}$** counting

mult: $f=a_n \prod (x-x_i)$.

Worked $x^4+16=0$: no real roots; 4 in $\mathbb{C}$; two:

$\sqrt{2}+i\sqrt{2}, -\sqrt{2}+i\sqrt{2}$ (**Check** $(\sqrt{2}+i\sqrt{2})^2=4i$,

$^2=-16$). All four: $\pm\sqrt{2} \pm i\sqrt{2}$ (signs independent;

$|x|=2$, angles $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$).

Proof (no real roots): for real x , $x^4 \geq 0$ (even power), so $x^4+16 \geq 16 > 0$ — never 0. I.e. $x^4=-16$ impossible over $\mathbb{R}$ (LHS ≥ 0). Hence all 4 FTA roots are nonreal. Factor thm:

$f(a)=0 \iff (x-a) \mid f$; $(x-1)^2=x^2-2x+1$

(multiple root); $x^2+1=0$: roots $\pm i$ (none real).

Note: 5 prime in $\mathbb{Z}$ but $5=(2-i)(2+i)$ in $\mathbb{Z}[i]$.

Coeff ($a_0 \dots a_n$); **value:** $\deg \leq n$ unique from $n+1$

pts (Vandermonde $\det \neq 0$). **Product $\deg n+m$**

needs $n+m+1$ pts; pad zeros

$(4+5x \rightarrow 4+5x+0x^2)$. Pointwise

$h(x_i)=f(x_i)g(x_i)$, $O(m)$; schoolbook $O(n^2)$.

Worked $f=1-x, g=2+x$ at $-2, 0, 1$:

$f: 1-(-2)=3, 1, 0$; $g: 2+(-2)=0, 2, 3$;

$h: 0, 2, 0 = h=2-x-x^2$ (**Check** $2+2-4=0, 2$,

$2-1-1=0$).

Sketch recipe: mark roots $\rightarrow$ lead-coeff sign for ends $\rightarrow y$ -intercept $\rightarrow$ join; graphical mult: sample ≥ 6 x 's, multiply y 's, plot.

Point-count recipe: single $\deg-n \rightarrow n+1$ pts; product $\deg n+m \rightarrow n+m+1$; FFT $\rightarrow$ next pow2 $\geq$ that.

7 FFT/IFFT + Costs

Even/odd: $p(x)=p_e(x^2)+x p_o(x^2)$. **Worked**

$1-2x+3x^2-4x^3+5x^4-6x^5+7x^6-8x^7$:

$p_e(y)=1+3y+5y^2+7y^3$, $p_o(y)=-2-4y-6y^2-8y^3$;

depth of recursion tree $=3$ ($8=2^3$).

Butterfly ($k=0..n/2-1$):

$p(\omega_n^k)=p_e(\omega_n^{2k})+\omega_n^k p_o(\omega_n^{2k})$;

$p(\omega_n^{k+n/2})=p_e(\omega_n^{2k})-\omega_n^k p_o(\omega_n^{2k})$. Needs $n=2^t$.

Butterfly $n=4$ ($\omega_4=i$):

$k=0$: $p(1)=p_e(1)+p_o(1)$

$p(-1)=p_e(1)-p_o(1)$

$k=1$: $p(i)=p_e(-1)+i p_o(-1)$

$p(-i)=p_e(-1)-i p_o(-1)$

NFFT $\geq 2n-1$, smallest pow2 (4+4 coeffs $\deg 3$:

product $\deg 6$, 7 coeffs $\Rightarrow \text{N}=8$). Pipeline:

coeff $\rightarrow$ FFT $\rightarrow$ vals $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ coeff.

Zero-pad; extra points OK if same poly. **Idea:** one

(p_e, p_o) pair gives two values (halves work).

IFFT: $a_j=\frac{1}{n} \sum_{k=0}^{n-1} y_k (\omega_n^{-j})^k$, $j=0..n-1$.

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$(i \cdot (-i) = 1)$.
 $a_2 = (y_0 + y_1(-1) + y_2 + y_3(-1))/4$
 $= (-1 - i + 1 - 2)/4 = (-2 - i)/4 = -0.5 - 0.25i$.
 $a_3 = (y_0 + y_1(i) + y_2(-1) + y_3(-i))/4$
 $= (-1 - 1 - 1 - 2i)/4 = (-3 - 2i)/4 = -0.75 - 0.5i$
 $(i \cdot i = -1)$.
Check $p(i) = a_0 + a_1 i - a_2 - a_3 i = i = y_1$.
Write p : $p(x) = (0.5 + 0.25i) + (-0.25 + 0.5i)x$
 $+ (-0.5 - 0.25i)x^2 + (-0.75 - 0.5i)x^3$.

8 Int Mult, GCD, Systems

Int as poly base B : $325 \rightarrow 5 + 2x + 3x^2$ ($p(10) = 325$).
 Steps: polys $\rightarrow$ FFT $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ carry to $[0, B-1] \rightarrow$ eval B .
Carry: $[2, 15, 7] \rightarrow [2, 5, 8] = 852$ (raw $2 + 150 + 700 = 852$; notes type 825).
 $5678 \rightarrow 8 + 7x + 6x^2 + 5x^3$; $q(10) = 5678$, $q(2) = 86$ (digits < 10 , no carry needed).
 d decimals $\rightarrow \lceil d/3 \rceil$ base-1000 digits: 120!:
 $199 \rightarrow 67$ ($66 \cdot 3 = 198 < 199$). **Idea:** every digit must land in $[0, B-1]$ (carry the rest).

GCD monic (lead 1; unique up to $\times \text{const}$):
 $4x^2 + 12x - 7 \rightarrow x^2 + 3x - 7/4$; $\gcd(x^2 - 1, x - 1) = x - 1$ ($((x-1)(x+1))$). Euclid via synthetic division.
Worked $f = x^4 - 8x^3 + 17x^2 + 2x - 24$,
 $g = x^3 - 6x^2 + 11x - 6$:

Step 1 (divide, two terms): $x^4/x^3 = x$, then
 $-2x^3/x^3 = -2$: $f - (x-2)g = -6x^2 + 30x - 36 = -6h$,
 $h = x^2 - 5x + 6$.

Step 2: $g \text{ rem } h$ is 0 $\Rightarrow \gcd(f, g) = h = x^2 - 5x + 6$.
Synthetic $(x^3 - 6x^2 + 11x - 6)/(x - 1)$, coeffs

		1	-6	11	-6
1, -6, 11, -6:	1		1	-5	6
		1	-5	6	0

Quot $x^2 - 5x + 6$, rem 0 (so $x = 1$ root). Non-root demo ($x + 1$): 1, -7, 18, -24, rem $-24 \neq 0$ (so $x = -1$ not a root).

Overdetermined: $h = \gcd$; $\deg h = 0 \Rightarrow$ inconsistent ($x - 1 = 0, x + 1 = 0$); $\deg \geq 1 \Rightarrow$ common roots = roots of h . Mini: $x^2 - 1 = 0$ (± 1) + $x - 1 = 0$ (1) $\Rightarrow$ common $x = 1$. Lecture consistent pair: $x - 1 = 0, 2x - 2 = 0$ (same root 1, $h = x - 1$).
Worked: $x^2 - 11x + 30 = 0$: $25 - 55 + 30 = 0$,
 $36 - 66 + 30 = 0 \Rightarrow 5, 6$.
 $x^3 - 10x^2 + 27x - 18 = 0$: test 6:
 $216 - 360 + 162 - 18 = 0$; roots 1, 3, 6 $\Rightarrow$ common $x = 6$.
Resultant: det condition for common root (eliminates a variable). 2×2 :
 $\text{Res}(x - 1, x + 1) = (1)(1) - (-1)(1) = 2 \neq 0 \Rightarrow$ no common root.

9 Worked Solutions (Step-by-Step)

Extra Q&A (Quiz/HW Style)

FFT size Qs: p with n coeffs needs n points forth+back; FFT needs pow2 so pad.
 $n - 1 = 16 \Rightarrow n = 17$ (not pow2) $\Rightarrow$ **cannot use directly**, pad to 32. $(p(x))^2$: $\deg 2(n - 1)$, needs $2n - 1$ pts $\rightarrow$ next pow2.
IFFT $n = 2$, $y = (1, 1 + i)$: $x_0 = 1 = \omega_2^0$, $x_1 = -1 = \omega_2^1$;
 $a_0 = (y_0 + y_1)/2 = (2 + i)/2 = 1 + 0.5i$;
 $a_1 = (y_0 - y_1)/2 = -0.5i$. **Check:** $p(1) = 1$,
 $p(-1) = 1 + i$.
 Gen a_2 formula ($n = 4$): $a_2 = (y_0 - y_1 + y_2 - y_3)/4$
 $(\omega_4^{-2k} = 1, -1, 1, -1)$. **Idea:** IFFT = FFT with ω_n^{-1} , then $/n$ (**trap:** forgetting $/n$ or the -1).
 C_8^* products ($z_1 = 1, z_2 = -1, z_5 = (1 + i)/\sqrt{2}$,
 $z_6 = (-1 + i)/\sqrt{2}, z_7 = (1 - i)/\sqrt{2}, z_8 = (-1 - i)/\sqrt{2}$):
 $z_2 z_5 = (-1 - i)/\sqrt{2} = z_8$; $z_5 z_7 = 1 = z_1$; $z_6 z_8 = 1 = z_1$.
 Inverse of $(-1 - i)/\sqrt{2}$: $|z|^2 = 1$, inv
 $= \bar{z} = (-1 + i)/\sqrt{2} = z_6$.

Group tests: $2a + 7bi$ ($a, b \in \mathbb{Z}$) additive: closed,
 $e = 0$, inv $-z \Rightarrow$ **group**. $3a + 3bi$, $ab \neq 0$
 multiplicative: $1 = 3a + 3bi \Rightarrow a = 1/3 \notin \mathbb{Z} \Rightarrow$ **not** (no identity).

D3 triangle symmetries (HW09):
 $\{e, r, r^2, \ell_1, \ell_2, \ell_3\}$, $r = \text{rot } 120^\circ$, $\ell_i = \text{reflections}$;
 $r^3 = e$, $\ell_i^2 = e$; $\text{rot} \cdot \text{rot} = \text{rot}$, $\text{refl} \cdot \text{refl} = \text{rot}$, mixed =
 refl (track one vertex to name it: $r^2 \cdot \ell_1 = \ell_2$,
 $r \cdot \ell_1 = \ell_3$); order 6, non-abelian ($r\ell_1 \neq \ell_1 r$); $r^{-1} = r^2$,
 each ℓ_i self-inverse. ($b \cdot a$: apply a first.)
FFT $n = 32$: depth 5 ($2^5 = 32$); butterfly pairs
 $(\omega_{32}^0, \omega_{32}^{16})$, $(\omega_{32}^1, \omega_{32}^{17})$; even/odd parts $\deg 15$, split
 again $\rightarrow 4$ polys ($p_{ee}, p_{eo}, p_{oe}, p_{oo}$). C_n^+ = cyclic
 additive order n ($\cong \mathbb{Z}_n^+$), gens $[k]_n$ with

$\gcd(k, n) = 1$; chain $C_{2^m}^* \subset C_{2^{m+1}}^*$.
 $p(\omega_{32}^0) = p_e(1) + p_o(1)$
 $p(\omega_{32}^{16}) = p_e(1) - p_o(1)$
 $p(\omega_{32}^2) = p_e(\omega_{32}^2) + \omega_{32} p_o(\omega_{32}^2)$
 $p(\omega_{32}^{17}) = p_e(\omega_{32}^2) - \omega_{32} p_o(\omega_{32}^2)$
Inconsistent check: $\gcd(x - 1, x + 1) = 1$ ($\deg 0$) $\Rightarrow$
inconsistent $((x + 1) - (x - 1) = 2 \neq 0)$.

Mod + Groups/FLT Drills

Reduce first:
 $[125]_{17} + [241]_{17} = [366]_{17}$; $366 - 21 \cdot 17 = 366 - 357 = 9$.
 $[391]_{21} \cdot [457]_{21}$: $391 - 18 \cdot 21 = 13$, $457 - 21 \cdot 21 = 16$;
 $13 \cdot 16 = 208 - 9 \cdot 21 = 19$.
 $[94]_7$: $94 - 13 \cdot 7 = 94 - 91 = 3$.
 $([157]_{13} + [249]_{13}) \cdot [83]_{13}$: $157 - 12 \cdot 13 = 1$,
 $249 - 18 \cdot 13 = 249 - 234 = 15$; $15 - 13 = 2$; $1 + 2 = 3$;
 $83 - 6 \cdot 13 = 5$; $3 \cdot 5 = 15 - 13 = 2$.

HW08 reduce-first:
 $[123]_{17} + [245]_{17} = 11$ ($368 - 357$). $[391]_{23} \cdot [457]_{23} = 0$
 $(391 = 17 \cdot 23)$.
 $([156]_{11} + [247]_{11}) \cdot [83]_{11}$: $2 + 5 = 7$, $\times 6 = 42 - 33 = 9$.
 $[312]_{15} \cdot ([184]_{15} + [257]_{15})$: $12 \cdot (4 + 2) = 72 - 60 = 12$.

Find x :
 $143 \equiv x \pmod{13}$: $143 = 11 \cdot 13 \Rightarrow 0$
 $12345 \pmod{11}$: $11 \cdot 1122 = 12342 \Rightarrow 3$
 $54321 \pmod{7}$: $7 \cdot 7760 = 54320 \Rightarrow 1$
 $147 \pmod{13}$: $13 \cdot 11 = 143 \Rightarrow 4$

Find n :
 $24 \equiv 0 \pmod{n} \Rightarrow n | 24$: 1, 2, 3, 4, 6, 8, 12, 24.
 $15 \equiv 1 \pmod{n} \Rightarrow n | 14$: 1, 2, 7, 14.
 $124 \equiv 4 \pmod{n} \Rightarrow n | 120$, $n > 4$:
 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. $18 \equiv 0$,
 $0 < n \leq 18$: 1, 2, 3, 6, 9, 18.

Classes: mod 7: 7 classes $[0] \dots [6]$.
 $[2]_5 = \{\dots, -8, -3, 2, 7, 12, \dots\}$; $[4]_7 = \{\dots, -3, 4, 11, \dots\}$;
 $[-3]_8 = [5]_8 = \{\dots, -3, 5, 13, \dots\}$. $\{1, -1\} \subset C_4^*$
 subgroup (closed, $e = 1$, self-inv).

Notation $\leftrightarrow$:
 $[8]_{11} + [9]_{11} = [6]_{11} \iff 8 + 9 \equiv 6 \pmod{11}$
 $(17 - 11 = 6)$; $[4]_7 \cdot [5]_7 = [6]_7 \iff 4 \cdot 5 \equiv 6 \pmod{7}$
 $(20 - 14 = 6)$.
 $17 + 9 \equiv 2 \pmod{8} \iff [17]_8 + [9]_8 = [2]_8$
 $(26 - 24 = 2)$; $6 \cdot 8 \equiv 3 \pmod{5} \iff [6]_5 \cdot [8]_5 = [3]_5$
 $(48 - 45 = 3)$.

Groups: $(\mathbb{Q}, +)$ group; $(\mathbb{Q}, \cdot)$ **not** (0 no inverse).
 $(\mathbb{Z}_4, +)$ group ($e = [0]$, inv $[-a]$). $\mathbb{Z}_4^* = \{1, 2, 3\}$ **not**
 $(2 \cdot 2 = 0)$.

Multi groups — which n admit one? $n = 3, 5, 13, 17$
 yes ($|G| = 2, 4, 12, 16$); 10, 15 no. Find gens: $\mathbb{Z}_5^*$ gens
 $\{2, 3\}$ (ord=4).

Possible orders (divisors of $|G|$): $2 \rightarrow \{1, 2\}$;
 $4 \rightarrow \{1, 2, 4\}$; $12 \rightarrow \{1, 2, 3, 4, 6, 12\}$;
 $16 \rightarrow \{1, 2, 4, 8, 16\}$. $|\mathbb{Z}_{24}^*| = \varphi(24) = 8$ (odd, not mult
 of 3: 1, 5, 7, 11, 13, 17, 19, 23); x odd (e.g. 3, 5, 7)
 never gives $[a]^x = [1]$ for $a = 2 \dots 23$ (needs $2|x$;
 $[5]^2 = [1]$ so every even x works).

Gen of $\mathbb{Z}_{13}^*$: $|G| = 12$, orders $\in \{1, 2, 3, 4, 6, 12\}$;
 $2^2 = 4$, $2^3 = 8$, $2^4 = 3$, $2^6 = 12 \neq 1 \Rightarrow \text{ord}(2) = 12 \Rightarrow$
 generator.

FLT big powers: $19^6 \pmod{7}$ OK ($7 \nmid 19$); 21^6 **no**
 $(\gcd = 7)$; $11^8 \pmod{9}$ **no** (9 composite).
 19^{68} : $19 \equiv 5$, $68 = 6 \cdot 11 + 2$ ($66 + 2$), $5^2 = 25 \equiv 4$.
 $271^{24} \pmod{11}$: $271 - 24 \cdot 11 = 271 - 264 = 7$.
 $24 = 10 \cdot 2 + 4$, $7^4 = 2401 - 218 \cdot 11 = 2401 - 2398 = 3$.
 $290^{293} \pmod{23}$: $290 - 12 \cdot 23 = 290 - 276 = 14$,
 $293 = 22 \cdot 13 + 7$ ($286 + 7$), $14^7 \equiv 19$.

Simplify without FLT: $[16^4]_7 = [2]$; $[10^{100}]_9 = [1]$;
 $[23^2]_9 = [7]$ ($23 \equiv 5$, $25 \equiv 18$); $[42]_{12} = [6]$.

More Euler: $5^{100} \pmod{13}$: $\varphi = 12$; $100 = 12 \cdot 8 + 4$;
 $5^4 = 625 - 48 \cdot 13 = 625 - 624 = 1$. $2^{1000} \pmod{7}$:
 $1000 = 6 \cdot 166 + 4$; $2^4 = 16 \equiv 2$.

Compute φ : $\varphi(8) = 8 - 4 = 4$;
 $\varphi(24) = \varphi(8)\varphi(3) = 4 \cdot 2 = 8$;
 $\varphi(100) = \varphi(4)\varphi(25) = 2 \cdot 20 = 40$. **Trap:**
 $\varphi(4)\varphi(2) = 2 \cdot 1 = 2 \neq 4 = \varphi(8)$ ($\gcd(4, 2) \neq 1$).
Euler trap: $2^4 \pmod{8} = 0 \neq 1$ though $\varphi(8) = 4$
 $(\gcd \neq 1)$.

Mod $x^3 - 1$ (Quiz4 Q6): $x^3 \equiv 1$:
 $[2x^5 - 9x^3 + 5x + 9] = 2x^2 - 9 + 5x + 9 = [2x^2 + 5x]$.
 Square after reducing:
 $[(2x^2 + 5x)^2] = [4x^4 + 20x^3 + 25x^2]$; $x^4 \equiv x$,
 $x^3 \equiv 1 \Rightarrow [25x^2 + 4x + 20]$.

Complex + FFT Drills

Sums (plot tip-to-tail, check):

$(-2 + 4i) + (-3 - 7i) = -5 - 3i$
 $(-2 - i) + (2 + i) = 0$
 $(1 - 2i) + (1 + 2i) = 2$
 $|z|^2 = z\bar{z}$: $(-2 - 3i)$: $4 + 9 = 13$; $4i$: 16;
 $\bar{z} = 4i \Rightarrow z = -4i$, $|z|^2 = 16$.
Div: $(2 + 6i)/2 = 1 + 3i$.
 $(3 + 4i)/(1 - 6i) = (3 + 4i)(1 + 6i)/37 = (-21 + 22i)/37$.
 $\approx -0.5676 + 0.5946i$ **Check** $(37 \cdot (-0.5676) = -21$,
 $37 \cdot 0.5946 = 22)$. **vs dot:** $z_1 z_2 = (ac - bd) + (ad + bc)i$
 vs $u \cdot v = ac + bd$; $|z_1 z_2|^2 = (a^2 + b^2)(c^2 + d^2)$ vs
 $(u \cdot v)^2 = (ac + bd)^2$; coincide $\iff ad = bc$.

E.g. $z_1 = 1 + 2i$, $z_2 = 3 + 4i$:
 $|z_1 z_2|^2 = (1 + 4)(9 + 16) = 125$ vs $(3 + 8)^2 = 121$
 $(1 \cdot 4 \neq 2 \cdot 3)$.

Even/odd twice of $p_e = 1 + 3y + 5y^2 + 7y^3$,
 $p_o = -2 - 4y - 6y^2 - 8y^3$:
 $p_{ee}(z) = 1 + 5z$, $p_{eo}(z) = 3 + 7z$
 $p_{oe}(z) = -2 - 6z$, $p_{oo}(z) = -4 - 8z$
Big-O tree (n leaves, balanced binary): depth
 $\log_2 n$; bottom-only work $\Rightarrow O(n)$; $O(n)$ per level
 $\Rightarrow O(n \log n)$.
 C_5^* row via iso:
 $\omega_5^2 \cdot \omega_5^2 = \omega_5^4 = \omega_5^3 \leftrightarrow [4] + [4] = [8] = [3]$. Gen $\omega_5 \leftrightarrow [1]_5$
 powers fill table.

Tables + FFT Drills

Mod-8 $\times$ table (nonzero rows): **Rule:** row a
 has 1 $\iff a$ invertible; row a hits 0 $\Rightarrow$ zero
 divisor. Units = odd $\{1, 3, 5, 7\}$ ($\gcd = 1$); $1^{-1} = 1$,
 $3^{-1} = 3$ ($9 - 8 = 1$), $5^{-1} = 5$ ($25 - 24 = 1$), $7^{-1} = 7$
 $(49 - 48 = 1)$. Even rows lack 1. **Zero divisors:**
 $2 \cdot 4 = 8 \equiv 0$, $6 \cdot 4 = 24 \equiv 0$ ($24 - 24$).

FFT $f = 2 + 3x + 5x^3$ ($N = 4$, $\omega_4 = i$; $p_e(y) = 2$,
 $p_o(y) = 3 + 5y$):
 $f(1) = 2 + 3 + 5 = 10$; $f(i) = 2 + 3i + 5(-i) = 2 - 2i$.
 $f(-1) = 2 - 3 - 5 = -6$; $f(-i) = 2 - 3i + 5i = 2 + 2i$.
Butterfly table (f): $k = 0$: $p_e(1) = 2$,
 $p_o(1) = 3 \Rightarrow f(1) = 2 + 8 = 10$, $f(-1) = 2 - 8 = -6$.
 $k = 1$: $p_e(-1) = 2$,
 $p_o(-1) = -2 \Rightarrow f(i) = 2 + i(-2) = 2 - 2i$,
 $f(-i) = 2 - i(-2) = 2 + 2i$.

DeMoivre power:
 $((1 + i)/\sqrt{2})^8 = (e^{i\pi/4})^8 = e^{2\pi i} = 1$ (angle $\times 8$).
 $((1 - i)/\sqrt{2})^2 = -i$, $((1 - i)/\sqrt{2})^4 = -1$ (angle
 $\times 4 = -\pi$). **Decrypt proof:** $m \equiv$
 $c^d = (m^b)^d = m^{bd} = m^{1+k(p-1)} = m \cdot (m^{p-1})^k \equiv m$
 (FLT, p prime, $p \nmid m$).

Tables + Step-by-Step Recoveries

Mod-8 $\times$ table (read inverses/zero divisors; odd
 rows permute, even rows hit 0):

$\times$	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

C_4^* vs $\mathbb{Z}_4^+$ (same shape $\Rightarrow$ iso $\omega_4^k \leftrightarrow [k]_4$):

*	1	i	-1	$-i$	+	0	1	2	3
1	1	i	-1	$-i$	0	0	1	2	3
i	i	-1	$-i$	1	1	1	2	3	0
-1	-1	$-i$	1	i	2	2	3	0	1
$-i$	$-i$	1	i	-1	3	3	0	1	2

C_5^* table via iso ($\omega_5^i \cdot \omega_5^j = \omega_5^{(i+j) \bmod 5}$):

*	1	ω_5	ω_5^2	ω_5^3	ω_5^4
1	1	ω_5	ω_5^2	ω_5^3	ω_5^4
ω_5	ω_5	ω_5^2	ω_5^3	ω_5^4	1
ω_5^2	ω_5^2	ω_5^3	ω_5^4	1	ω_5
ω_5^3	ω_5^3	ω_5^4	1	ω_5	ω_5^2
ω_5^4	ω_5^4	1	ω_5	ω_5^2	ω_5^3

Recover $h = a + bx + cx^2$ from values $(0, 2, 0)$ at
 $(-2, 0, 1)$: $a - 2b + 4c = 0$, $a = 2$, $a + b + c = 0$.

Step 1: $a = 2$. **Step 2:** $b + c = -2$,
 $-2b + 4c = -2 \Rightarrow b - 2c = 1$.

Step 3: subtract: $3c = -3 \Rightarrow c = -1$, $b = -1$. **Box:**
 $h = 2 - x - x^2$. **Check:** $2 + 2 - 4 = 0$, $2, 2 - 1 - 1 = 0$.

Why unique: Vandermonde det at $-2, 0, 1$:
 $(0 + 2)(1 + 2)(1 - 0) = 2 \cdot 3 \cdot 1 = 6 \neq 0$. **Read graph $\rightarrow$**
number: point $(x, y) \rightarrow x + yi$; length r , angle
 $\theta \rightarrow r(\cos \theta + i \sin \theta)$.